−2 fuel savings
−1 cost savings
−1 ROI

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Homework 3 - Mahya Asadalizadeh

ChE451

Heavy commercial vehicles like van trailers are aerodynamically inefficient due to their unstreamlined shapes, with significant air resistance leading to substantial fuel consumption. For a van trailer traveling at 100 km/h, approximately 53% of the total fuel is consumed to overcome this aerodynamic drag.

Aerodynamic designs aim to reduce air resistance (drag) and improve fuel economy. Even minor reductions in drag can greatly impact fuel efficiency.

The main devices used to reduce drag are trailer skirts and trailer tails.

- Trailer Skirts: are panels installed along the bottom sides of a trailer. They streamline the airflow beneath the trailer, reducing air turbulence and drag. They enhance fuel efficiency by minimizing the low-pressure zone (they improve fuel efficiency from 3% to 7% and if combined with other measures, savings can reach up to 15%).
- Trailer Tails: are aerodynamic devices mounted at the rear of a trailer and typically swing outward to extend the length of the trailer when deployed. They reduce the low-pressure area at the rear of the trailer, minimizing the drag caused by air turbulence (they improve fuel efficiency up to 5% - 7%).

❖ Using trailer tails and skirts together can provide better fuel savings (reducing fuel consumption up to 9%). These improvements save costs and reduce emissions, making them an environmentally friendly option for the trucking industry.

✓ Mathematics:

Aerodynamic drag is the major source of fuel consumption. The power required to overcome drag (P_d) is directly related to the drag coefficient (C_d) and the cube of the vehicle speed (V^3):

[illegible] $\times C_d \times A \times V^3$

P_d = Power required to overcome aerodynamic drag

ρ = Air density

C_d = Drag coefficient

A = Projected frontal area

V = Vehicle speed

The fuel consumed is proportional to the required work against the drag force. So, the force due to aerodynamic drag is:

[illegible] $\times C_d \times A \times V^3$

➢ The fuel consumed is proportional to the required work against the drag force. So, reducing the drag coefficient (C_d) directly reduces the amount of fuel that needs to be consumed ($\dot{m}_{fuel}$).